

Recommending Makerspace Best Practices Based On Visualization of Student Use Data

Jordan Phan and Jennifer Truong

University of Virginia

jordan.phan@virginia.edu / hcz3yb@virginia.edu



ISAM 2024

International
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Abstract

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Research Questions

1. What are the most common reasons that users visit the Makerspace?
2. How can we increase visits?
3. What are best practices to develop to support students' needs?



Relevance

The purpose of this paper is to utilize lab collected data to:

- Optimize **availability** for students
- Increase equipment and workshop **engagement**





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Introduction / Background

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UVA Scholars' Lab Makerspace

The UVA Scholars' Lab Makerspace is a space for students, staff, faculty, and community members to learn and make collaboratively.



UVA Scholars' Lab Makerspace

Some of the equipment in the lab:

- Arduinos
- Sewing Machines
- Button Makers
- Hand Tools
- Cameo
- Cintiq Tablet
- Power Tools
- 3D Printers
- Photogrammetry Equipment
- Soldering Irons

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Methodology + Data

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Data Collection Process

Visit check-in form

- Kiosk at the entrance of the Makerspace
- Name, email, UVA status, affiliation, reason for visit
- March 2022 - March 2024
- Equipment check-out

Visitors may reserve equipment through online form

- Technologists complete the form for walk-in visitors
- Date, time, availability of equipment, duration of use
- 2014 - 2024, with missingness before 2023

Limitations

- Missingness in data
 - **No enforcement mechanism for visit check-ins**
 - **Not all equipment requires check-out**
- 3D printers only recently available
- Laser cutter not reflected in data
- Observational data



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Findings

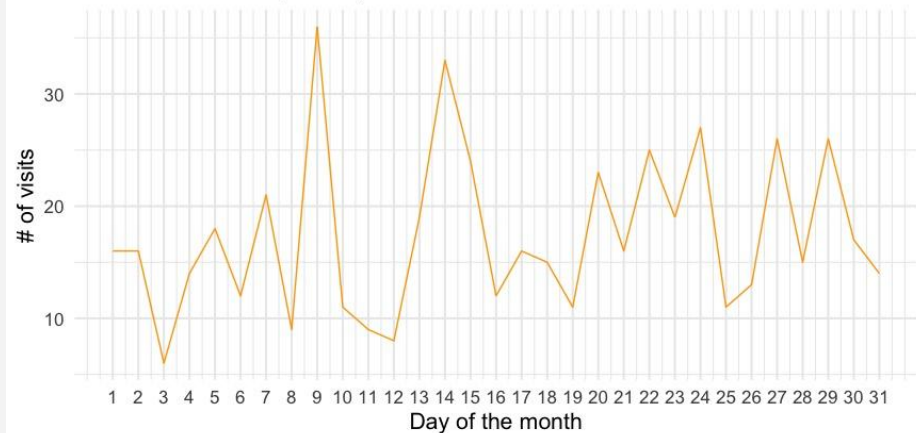
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Check-in Data

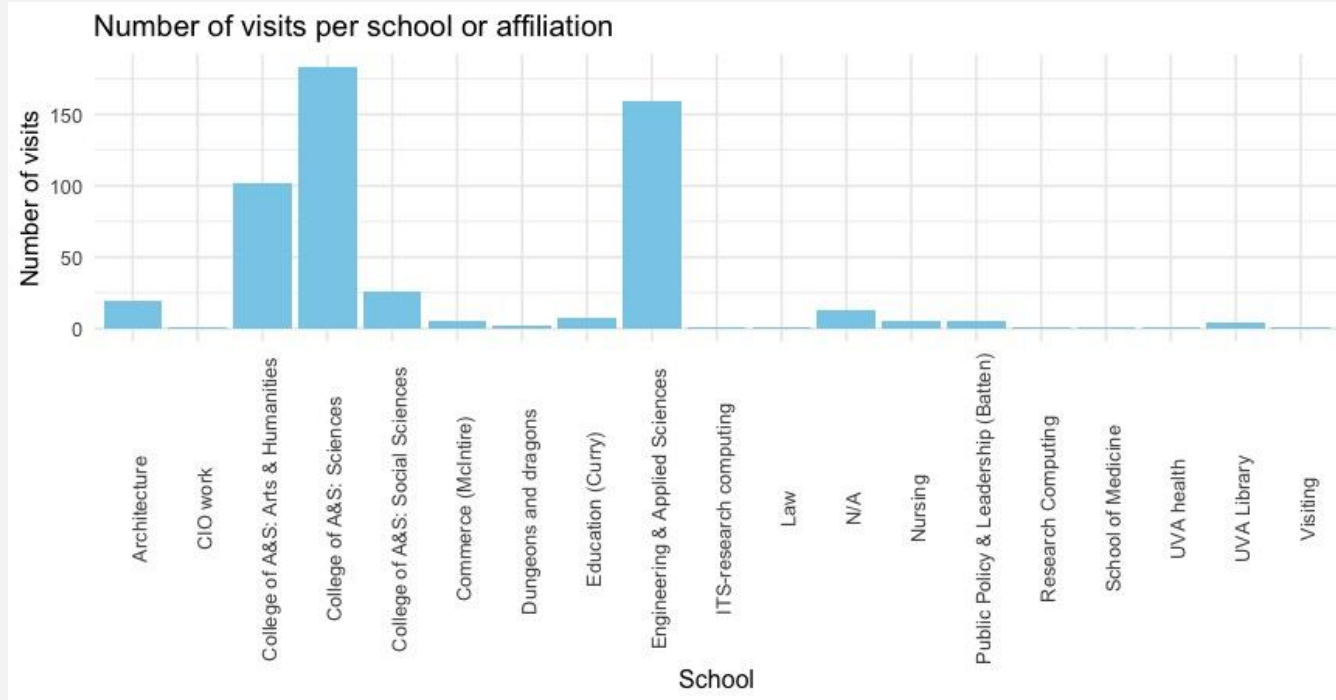
Number of visits per hour of the day



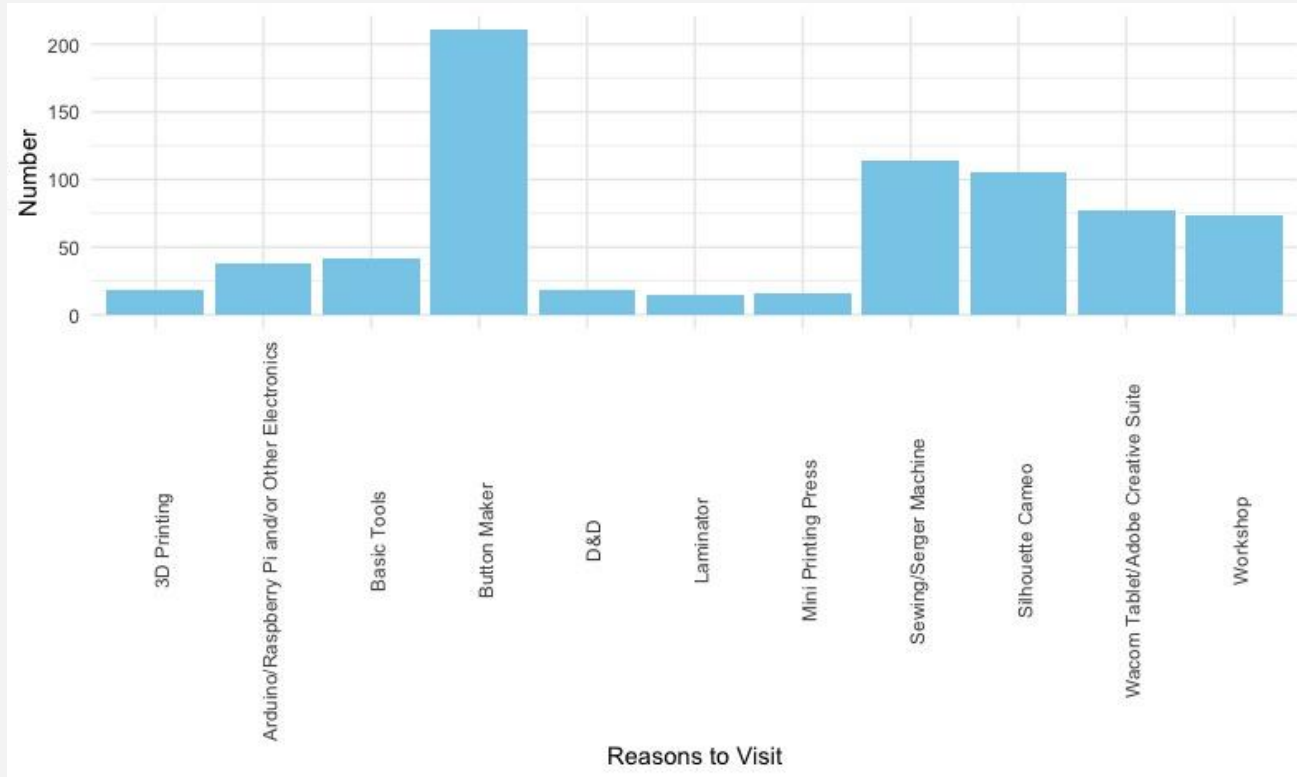
Number of visits per day of the month



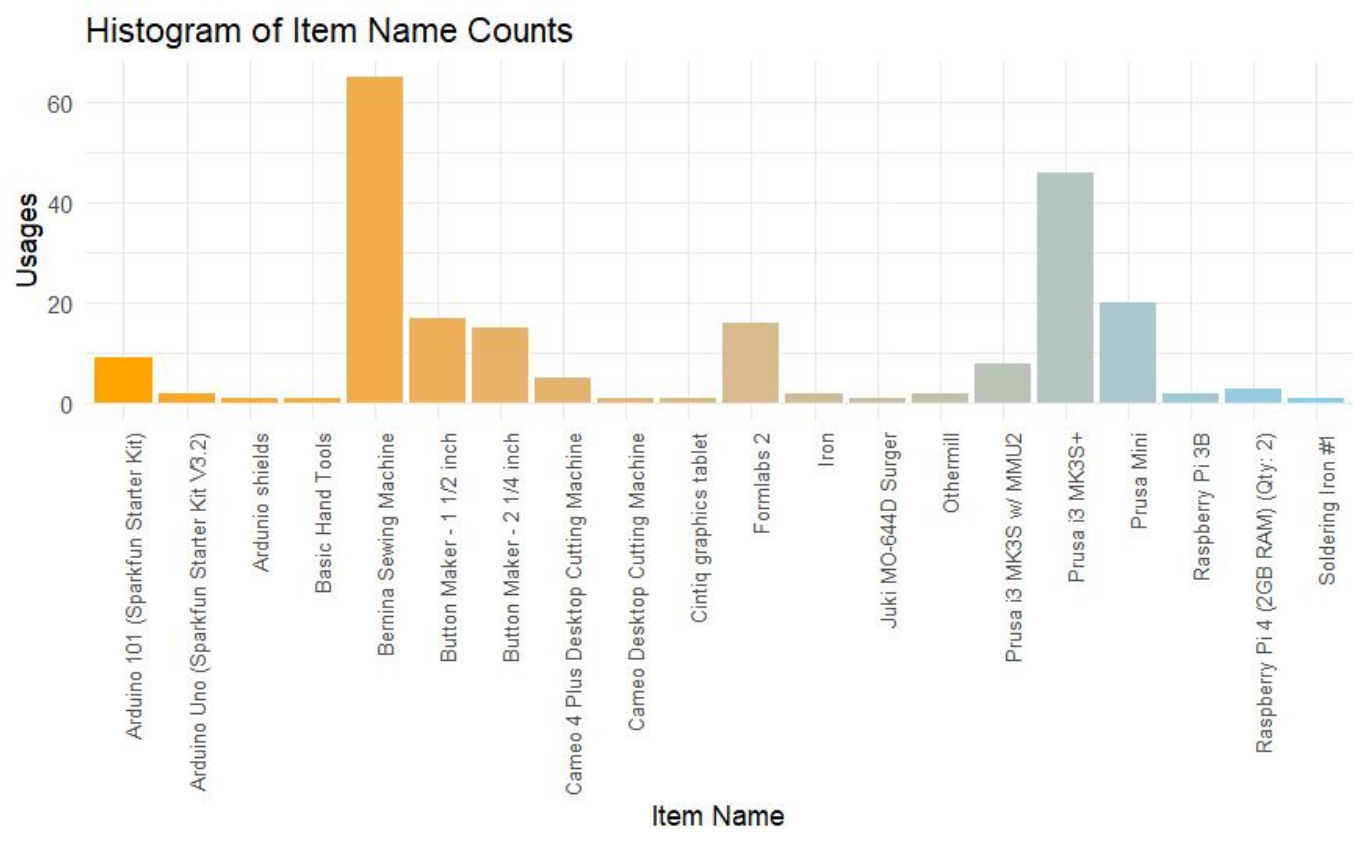
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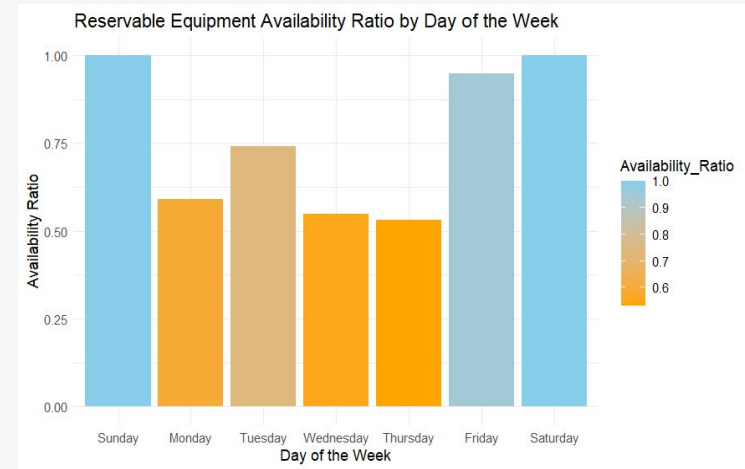
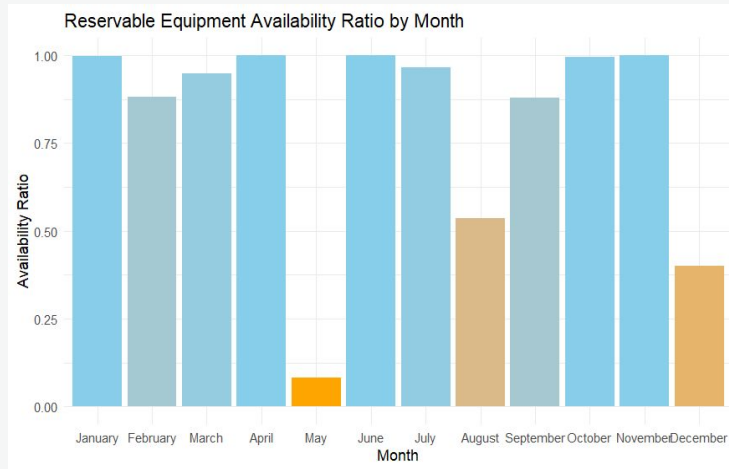
Check-in Data



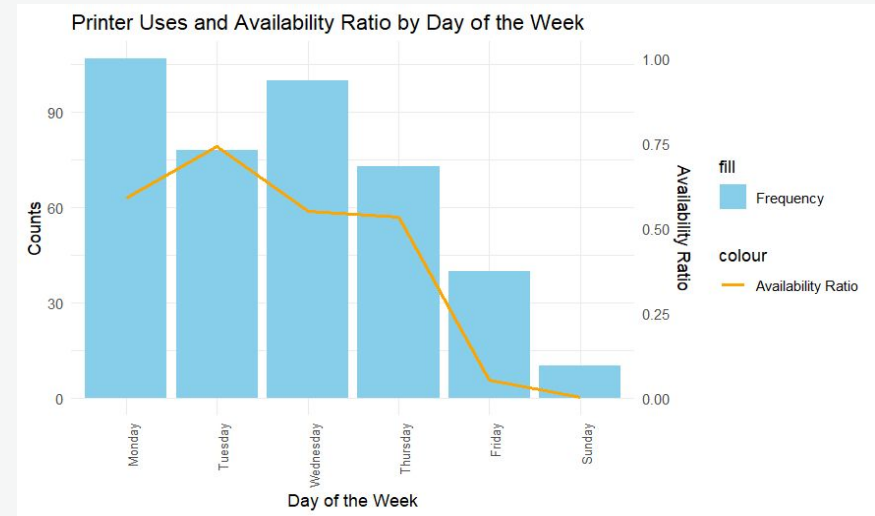
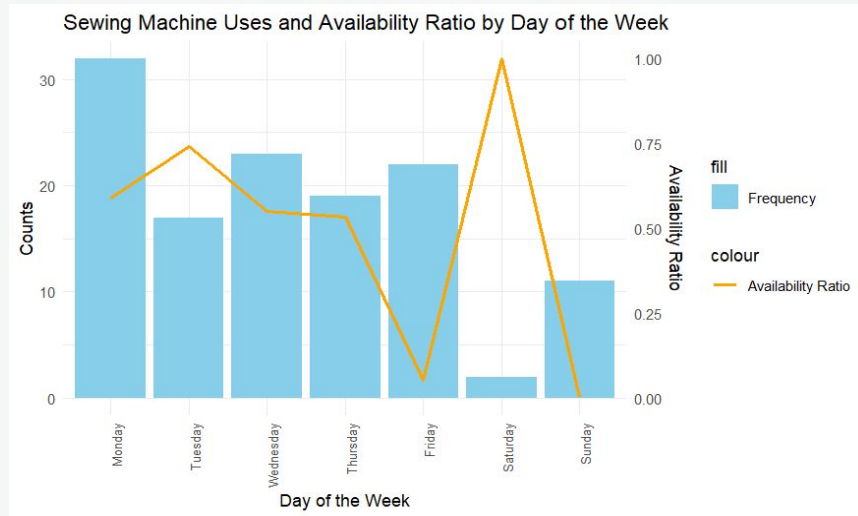
Equipment Usage



Equipment Availability



Equipment Availability (closer look)



Availability Ratio = 1 - Reserved Hours / Total Hours Available

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Conclusion

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Recommendations

- Proactively collect data across all users to address data limitations.
- Arrange open hours and publication of hours around high-use periods to better serve students.
- Increase visits through special events and low barrier-to-entry crafts.



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Thank you.

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